

1. Introduction & Literature Review

Explain the background of your project.

The intention of the project is to see the various factors that may impact. The average talk duration of a call. Whether it's the date of the week. the agent, the topic. Information like this can impact how. A setter performs overall. Our satisfied customers. Will be. And any staffing needs that need to be updated Some current companies like Nice IX, Calabrio, Aspect Workforce all have current workforce tools. To try to help forecast and anticipate. How they might perform on a given day All of these companies are currently using AI tools to help optimization using AI chat bots. To limit the need for additional staffing and shortened call times for the centers (Dubois, P. (2026, March 5 and (PR, 2025)). From personal experience and using NICE IEX uh, some of the UI is not unfriendly. A lot of it is very manual, you'd have to submit a certain requests, and it's not offered automatically.

2. Problem Definition

The plan is to help predict trends and average talk duration (my target) based on Agent, Topic Day of the week(input) This would be a linear regression model as and it takes into several different factors to get its outcome.

Goal: Make your problem precise and measurable.

3. Approach, Materials, and Methods

The data set I used was Call Center Statistics directly from GitHub.Call center statistics directly from GitHub. <https://github.com/globalsmile/Call-Center-Analysis/raw/main/01%20Call-Center-Dataset.xlsx> It is a set with 5000 rows and general information such as call ID agent, Date time stamp, Call duration topics and satisfaction scores. A lot of the EDA steps I used were just evaluating the structure. Of the columns. Changing the date to days of the week. Removing unneeded columns like call ID. Timestamps.

- Models used (e.g., Logistic Regression, Random Forest, Neural Network)

- Why you selected these models

I chose linear regression because I wanted to get numerical information from the average talk duration. And I also used logistics for the categories. I tried several different means using the R-squared MAE RMSE. I used Python SK Learn.

4. Results

I had many failures I think I used the features in the wrong way, encoded in the wrong place I hope to improve with the next one I ended up just using avg talk duration in seconds using the mean and the topic.

5. Discussion

Explain what your results mean.

My model did not perform well. I feel like some lot of the encoding ended up um. Not being useful for the model and. Not getting remotely close to what was needed. Like as in I use too many features at a time. Tried several different iterations, several different ways. In the end I just picked a very simple model of an average regression advtalkduration per call topic.

6. Conclusion

Summarize your work.

While using AI tools, I learned several different ways of handling and prepping the data. As well as alternative. Plots and charts I could use in the future. More practice is needed for me to improve. This exercise was great to learn more about the EDA process and how to prep data and what not to do when building these models.

Citations

Dubois, P. (2026, March 5). 26 call center statistics that reveal where AI is actually working in 2026. CMSWire.com. <https://www.cmswire.com/contact-center/16-important-call-center-statistics-to-know-about/>

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PR, T. (2025, March 25). 98% of contact centers are using AI, and 61% are experiencing more difficult conversations, according to New Calabrio Research. Global Press Release & Newswire Distribution Services. <https://www.businesswire.com/news/home/20250325095245/en/98-of-Contact-Centers-Are-Using-AI-and-61-Are-Experiencing-More-Difficult-Conversations-According-to-New-Calabrio-Research>